OFFICIAL TECHNICAL BLUEPRINT · V2.1

QuickFinSure IT System Architecture & Flowcharts

Complete multi-stakeholder operational swimlanes, loan lifecycle state machine, cloud infrastructure, Capacitor mobile synchronization, and relational database schema.

Platform: QuickFinSure Solutions **Live URL:** https://quicfinsure.onrender.com **Target Audience:** IT, Engineering & Operations

Date: September 2026

This document serves as the official technical manual and blueprint for the **IT, Engineering, DevOps, and Operations teams**. It details the cloud infrastructure, lifecycle state machine, operational swimlane workflows, sequence interactions, data flow, and database schema.

1. High-Level System & Cloud Architecture



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2. Loan Origination State Machine (Status Lifecycle)

The QuickFinSure core engine tracks every loan application through an explicit state machine. Each state transition enforces specific validation rules, logs an immutable audit trail (`loan_status_history`), and triggers multi-channel alerts.



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3. End-to-End Operational Swimlane Flowchart

This diagram maps responsibilities across all stakeholders: **Channel Partner (DSA)**, **Customer (Borrower)**, **Branch Operations**, **Partner Bank**, and **Finance / Accounts**.



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4. Sequence Diagram: Tokenized Lead Capture & Document Pipeline

This diagram shows the exact technical communication, API calls, and database updates during the customer submission flow.



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5. Mobile App (Capacitor) Real-Time Synchronization Flow

Explains how the native Android APK operates with the cloud backend without requiring re-compilation for UI updates.



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6. Multi-Tenant White-Labeling Architecture

QuickFinSure supports multi-tenancy. Every tenant (DSA, NBFC, or white-label brokerage) can have independent corporate branding, colors, logos, and user hierarchies.



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7. Database Entity Relationship Diagram (ERD)



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8. IT Maintenance & Operational Runbook

Key Endpoints for DevOps & IT Health

Endpoint	Method	Purpose	Authorized Roles
`/api/auth/[...nextauth]`	`GET/POST`	Session validation, login, token refresh	Public
`/api/db-debug`	`GET`	Verifies DB connectivity (Neon / fallback)	Super Admin
`/apply/[token]`	`GET/POST`	Public Customer Application & Document Upload	Public (Tokenized)
`/card/[id]`	`GET/POST`	Public Advisor Digital Visiting Card & Lead Box	Public
`/dashboard`	`GET`	Role-based Dashboard & Metrics	Authenticated
`/branch/requests/[id]`	`GET/POST`	9-Stage Operational File Processor	Operations, Admin, RM
`/QuickFinSure.apk`	`GET`	Direct download of latest Android APK	Public

Environment Variables Matrix

Key	Required	Example / Default	Description
`DATABASE_URL`	**Yes**	`postgresql://user:pass@ep-xyz.neon.tech/neondb`	Primary Neon PostgreSQL connection string with SSL mode
`NEXTAUTH_SECRET`	**Yes**	`[32-byte-hex-secret]`	Encryption key for signing user sessions & JWTs
`NEXTAUTH_URL`	**Yes**	`https://quicfinsure.onrender.com`	Base URL used by NextAuth for callback validation
`APP_BASE_URL`	Optional	`https://quicfinsure.onrender.com`	Canonical public origin used for generating shareable links
`SMTP_HOST`	Optional	`smtp.gmail.com` / `smtp.sendgrid.net`	Host for transactional notifications
`SMTP_USER`	Optional	`notifications@quickfinsure.com`	SMTP authentication username
`SMTP_PASSWORD`	Optional	`[app-password]`	SMTP app password